DAY 26 | POST-READ

Agents and Agentic RAG: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Agent loop	Goal, plan, act, observe, reflect, repeat until done or stopped.
Components	Model, instructions, tools, memory, guardrails, orchestration.
Workflow vs agent	Fixed steps when you can; agents when the path depends on findings.
Agents SDK	Agent, @function_tool, Runner, handoffs, guardrails, tracing.
Agentic RAG	The agent decides whether, where and how to retrieve.
CRAG	Retrieve, grade, correct (rewrite or search elsewhere), generate.
Supervisor pattern	A coordinator routes to specialists with focused tools.
Multi-agent cost	Every hop adds latency, tokens and debugging effort.

Part B: Going beyond today

1. Agent memory

Short-term memory is the conversation and scratchpad. Long-term memory stores facts across sessions, usually in a database or vector store, retrieved only when relevant. Let users see and delete what is remembered.

2. Self-RAG and reflection

Beyond grading documents, agents can grade their own draft: "Is every claim supported by the evidence?" If not, revise. This costs extra calls but reduces unsupported answers.

3. Running the Agents SDK in notebooks

```
result = await Runner.run(agent, "Where is ORD-1042?")   # top-level await works in Colab and
print(result.final_output)
```

Runner.run_sync is convenient in scripts. In notebooks, which already run an event loop, use await Runner.run(...) or nest_asyncio.

4. Agent evaluation

Evaluate agents on outcomes (task solved?), process (sensible tool use, steps taken) and safety (no forbidden actions). You will build evaluations on Days 29 and 30.

5. Cost control

Cap steps and tokens per task, use smaller models for grading and routing, cache tool results, and log cost per run.

Part C: Check yourself

- What makes an agent different from a single LLM call?
- What is a handoff in the Agents SDK?
- What does the grading step in CRAG decide?
- When does a supervisor with specialists make sense?
- Name three ways to limit an agent's cost.

Part D: Practice ideas for tonight

- Add a reflection node to the CRAG graph that checks the answer against the evidence.
- Give the analyst agent a second dataset and a new question.
- With an API key, run the Agents SDK example and read its trace.

Coming up on Day 27: agent protocols (MCP and A2A), agentic handoff, a data analysis agent, and a senior FDE session.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today